🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.DS.6C

Mean Absolute Deviation

Lesson 8-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the mean absolute deviation (MAD) to describe how spread out data is.
Language Objective: I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean Absolute Deviation · **Deviation** · **Absolute Value** · **Spread** · **Data distribution** · **Variability**

👆 *Tap any word to see what it means and a picture.*

1. The average distance of each value from the mean is the ________ .

2. How far a single value is from the mean is its ________ .

3. The distance of a number from zero, always positive, is its ________ .

4. How far apart the data values are from each other is the ________ .

5. The way data values are spread out is the ________ .

6 How much the data values differ is the ________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

To find the MAD for Player A (18, 22, 20, 24, 16, mean 20), why do we take the ABSOLUTE VALUE of each deviation before averaging?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Mean Absolute Deviation** — The average distance of each number from the mean.
  Desviación media absoluta — La distancia promedio de cada número a la media.
- ☐ **Deviation** — How far a number is from the mean.
  Desviación — Qué tan lejos está un número de la media.
- ☐ **Absolute Value** — How far a number is from zero. It is always positive.
  Valor absoluto — Qué tan lejos está un número de cero. Siempre es positivo.
- ☐ **Spread** — How far apart the numbers are.
  Dispersión — Qué tan separados están los números.
- ☐ **Data distribution** — How the data looks: where it sits and how spread out it is.
  Distribución de datos — Cómo se ven los datos: dónde están y qué tan separados están.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

We use absolute value so the deviations are all ____.

Usamos el valor absoluto para que las desviaciones sean todas ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Two data sets can share a mean but differ in how spread out they are.

Evidence: Students choose Player A as more consistent and recognize that equal means hide different spreads.

Reasoning: This evidence connects the definition of mean absolute deviation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **We use absolute value so the deviations are all ____.**
  Usamos el valor absoluto para que las desviaciones sean todas ____.
- **Without absolute value, the deviations would add up to ____.**
  Sin valor absoluto, las desviaciones sumarían ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean Absolute Deviation
2. **Notes 2:** Deviation
3. **Notes 3:** Absolute Value
4. **Notes 4:** Spread
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Mean Absolute Deviation is adding the signed deviations (value − mean) instead of their absolute values, so the positives and negatives cancel out to 0. For 6, 8, 10, 12 (mean = 9), the deviations are −3, −1, 1, 3 — they sum to 0, but that is NOT the MAD. Before averaging, take the absolute value of every deviation so each becomes a positive distance: $|-3| = 3$, $|-1| = 1$, $|1| = 1$, $|3| = 3$. Sum = 8, so MAD = 8 ÷ 4 = 2. A MAD of exactly 0 is a warning sign — it only happens if every value in the data set equals the mean.*
8. **Try It 1:** B. 2 — *Add the distances: 3 + 1 + 1 + 3 = 8. Divide by the 4 games: 8 ÷ 4 = 2. The MAD is 2 points.*
9. **Try It 2:** A. Shooter A, because a smaller MAD means scores stay closer to the average — *A smaller MAD means the scores hug the mean. Shooter A's MAD of 2 is smaller than Shooter B's 6, so Shooter A is more consistent.*